# Explicit lower bounds for runs of three consecutive quintic and septic residues

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**Abstract**

For a prime $p$ let $r(k,3,p)$ be the least $r \geq 1$ such that $r, r+1, r+2$ are all $k$-th power residues modulo $p$, and $\Lambda(k,3) = \limsup_{p\to\infty} r(k,3,p)$. Lehmer, Lehmer, Mills and Selfridge proved $\Lambda(3,3) = 23532$; whether $\Lambda(k,3)$ is finite for odd $k \geq 5$ is open (Erdős Problem #436). We give explicit assignments of quintic (resp. septic) characters to the primes and prove, with a Lean 4 certificate (one `native_decide` evaluation, everything else kernel-checked), that no completely additive function realising them has three consecutive zeros below $10^8$ (resp. $10^7$). Combined with the classical Kummer–Chebotarev realisability argument, which we prove in the paper but did not formalise, this gives $\Lambda(5,3) \geq 99\,999\,999$ and $\Lambda(7,3) \geq 9\,999\,999$. Computations outside Lean show that the same assignments have their first triples of consecutive zeros at $313\,714\,024$ and $180\,317\,500$, so $\Lambda(5,3) \geq 313\,714\,024$ and $\Lambda(7,3) \geq 180\,317\,500$ (computational, not formally certified). The bounds previously available were the trivial $\Lambda(5,3) \geq \Lambda(5,2) = 7888$ and $\Lambda(7,3) \geq \Lambda(7,2) = 1\,649\,375$. Whether $\Lambda(5,3)$ and $\Lambda(7,3)$ are finite remains open.

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# 1 Introduction

Let $k, m \geq 2$. By a theorem of Brauer [3], every sufficiently large prime $p$ has $m$ consecutive positive integers $r, r+1, \ldots, r+m-1$ that are all $k$-th power residues modulo $p$; let $r(k,m,p)$ denote the least such $r$ and put

$$\Lambda(k,m) = \limsup_{p\to\infty} r(k,m,p) \in \mathbb{N} \cup \{\infty\}.$$

D. H. Lehmer and E. Lehmer [12] started the study of $\Lambda$: they observed $\Lambda(2,2) = 9$ (if neither 2 nor 5 is a quadratic residue then 10 is), and proved $\Lambda(2,3) = \infty$ and $\Lambda(k,4) = \infty$ for $k \leq 1048909$; more generally $\Lambda(k,m) = \infty$ whenever $k$ is even and $m \geq 3$ [12, 13], and Graham [8] showed $\Lambda(k,m) = \infty$ for all $k$ and all $m \geq 4$. The finite values known are

$$\Lambda(3,2) = 77 \text{ [6]}, \qquad \Lambda(4,2) = 1224 \text{ [1]}, \qquad \Lambda(5,2) = 7888, \quad \Lambda(6,2) = 202124 \text{ [13]},$$
$$\Lambda(7,2) = 1649375 \text{ [4]},$$

and, for runs of length three, the "machine proof"

$$\Lambda(3,3) = 23532 \qquad \text{(Lehmer, Lehmer, Mills and Selfridge [14])}.$$

Hildebrand proved that $\Lambda(k,2) < \infty$ for every prime $k$ [10] and then for every $k$ [11] (a combinatorial proof via a generalised Schur theorem is given by Dietzel [5]). Brillhart, Lehmer and Lehmer [4] also proved the general lower bound $\Lambda(k,2) > 5^k$ for every odd $k \geq 5$ (and for $k$ twice an odd prime $\equiv 3 \pmod 4$).

Erdős and Graham [7, p. 87] asked whether $\Lambda(k,3)$ is finite for every odd $k$ and how large $\Lambda(k,2)$ and $\Lambda(k,3)$ are; this is Problem #436 on the Erdős problems website [2], which (page last edited 25 October 2025, consulted October 8, 2026) records as the remaining questions "whether $\Lambda(k,3)$ is finite for all odd $k \geq 5$, and the growth rate of $\Lambda(k,2)$ and $\Lambda(k,3)$ as functions of $k$". The first open case is $k = 5$.

**What is new.** Every run of three residues begins with a pair, so $r(k,3,p) \geq r(k,2,p)$ and trivially $\Lambda(5,3) \geq \Lambda(5,2) = 7888$, $\Lambda(7,3) \geq \Lambda(7,2) = 1\,649\,375$ (and $\Lambda(k,3) \geq \Lambda(k,2) > 5^k$ for every odd $k \geq 5$ by [4]; the $\Omega$-assignment of Section 5 gives $\Lambda(k,3) \geq 3^k - 2$). To our knowledge, as of October 8, 2026, no bound specific to runs of *three* quintic or septic residues, i.e. going beyond these trivial consequences of the pair results, has been published. We give the first such bounds:

**Theorem 1.1** (main result)**.**

$$\Lambda(5,3) \;\geq\; 99\,999\,999 \qquad \textit{and} \qquad \Lambda(7,3) \;\geq\; 9\,999\,999.$$

*More precisely, for the explicit prime assignments $g_5, g_7$ of Section 3, there are infinitely many primes $P \equiv 1 \pmod 5$ (resp. $P \equiv 1 \pmod 7$) with $r(5,3,P) \geq 10^8 - 1$ (resp. $r(7,3,P) \geq 10^7 - 1$).*

**Logical structure.** The proof has two parts of different status.

1. A purely combinatorial statement about completely additive functions with prescribed prime values (Theorem 4.1). It is *certified in Lean 4* (core library only, no Mathlib). The certificate uses `native_decide` exactly once, to evaluate one large finite check on a table of $10^8 + 1$ bytes; everything else is checked by the Lean kernel (Section 4).

2. The passage from completely additive functions to $k$-th power residues modulo infinitely many primes $P$: a classical consequence of Kummer theory and the Chebotarev density theorem (Lemma 2.2). We give a complete proof but have *not* formalised it.

In addition we report stronger bounds that are *computational only* (Section 5):

$$\Lambda(5,3) \;\geq\; 313\,714\,024, \qquad \Lambda(7,3) \;\geq\; 180\,317\,500.$$

**What is not shown.** We do not know whether $\Lambda(5,3)$ or $\Lambda(7,3)$ is finite. Our searches found no indication of a threshold beyond which every assignment of characters produces a zero triple (Section 6); if $\Lambda(5,3)$ is finite, it exceeds $3 \cdot 10^8$, about 13 000 times $\Lambda(3,3)$.

## 2 Reduction to completely additive functions

Throughout, $k$ is an odd prime. For a prime $P \equiv 1 \pmod{k}$ the $k$-th power residues modulo $P$ form the subgroup of index $k$ of the cyclic group $\mathbb{F}_P^\times$. Fix a primitive $k$-th root of unity $\zeta = \zeta_k$ in $K = \mathbb{Q}(\zeta)$.

**Definition 2.1.** An *assignment* is a map $g$ from the primes to $\mathbb{Z}/k\mathbb{Z}$. A function $f : \mathbb{N} \to \mathbb{N}$ is *completely additive mod $k$ on* $[1, L]$ if $f(ab) \equiv f(a) + f(b) \pmod{k}$ whenever $a, b \geq 1$ and $ab \leq L$; it *realises $g$ on* $[1, L]$ if moreover $f(q) \equiv g(q) \pmod{k}$ for every prime $q \leq L$. (Such $f$ are unique mod $k$ on $[1, L]$; see the proof of Theorem 4.1.)

**Lemma 2.2** (realisability)**.** *Let $k$ be an odd prime, $q_1 < \cdots < q_t$ distinct primes ($k$ itself may be among them) and $c \in (\mathbb{Z}/k\mathbb{Z})^t$. There are infinitely many primes $P \equiv 1 \pmod{k}$ for which there is a map $f_P : \{n \in \mathbb{Z} : P \nmid n\} \to \mathbb{Z}/k\mathbb{Z}$ such that*

1. *$f_P(mn) = f_P(m) + f_P(n)$;*
2. *$f_P(q_i) = c_i$ for $i = 1, \ldots, t$;*
3. *$n$ is a $k$-th power residue modulo $P$ if and only if $f_P(n) = 0$.*

*Proof. Independence.* Let $K = \mathbb{Q}(\zeta)$, so $[K : \mathbb{Q}] = k - 1$ and $\mu_k \subset K$. Suppose $x = \prod_i q_i^{a_i} = \alpha^k$ with $\alpha \in K^\times$ and $0 \leq a_i < k$. Taking the norm $N_{K/\mathbb{Q}}$, and using that $x \in \mathbb{Q}$,

$$\prod_i q_i^{(k-1)a_i} = N_{K/\mathbb{Q}}(x) = N_{K/\mathbb{Q}}(\alpha)^k \in (\mathbb{Q}^\times)^k.$$

Comparing $q_i$-adic valuations gives $k \mid (k-1)a_i$, hence $k \mid a_i$ and $a_i = 0$. So the classes of $q_1, \ldots, q_t$ are $\mathbb{F}_k$-linearly independent in $K^\times/(K^\times)^k$.

*Kummer theory.* Let $\Delta \subseteq K^\times$ be generated by $(K^\times)^k$ and $q_1, \ldots, q_t$, and $M = K(\Delta^{1/k}) = K(q_1^{1/k}, \ldots, q_t^{1/k})$ for fixed choices of the roots. Since $\mu_k \subset K$, Kummer theory gives $[M : K] = |\Delta/(K^\times)^k| = k^t$, and the map

$$\mathrm{Gal}(M/K) \to \mu_k^t, \qquad \sigma \mapsto \left(\sigma(q_i^{1/k})/q_i^{1/k}\right)_i$$

is an injective homomorphism between groups of order $k^t$, hence an isomorphism. Let $\sigma_c$ be the element with $\sigma_c(q_i^{1/k}) = \zeta^{c_i} q_i^{1/k}$ for all $i$.

*Chebotarev.* $M/K$ is abelian and unramified outside the primes dividing $kq_1 \cdots q_t$. By the Chebotarev density theorem the primes $\mathfrak{p}$ of $K$, unramified in $M$, with Frobenius $\mathrm{Frob}_\mathfrak{p} = \sigma_c$

have Dirichlet density $k^{-t} > 0$. Primes $\mathfrak{p}$ of residue degree $\geq 2$ over $\mathbb{Q}$ have density 0 (at most $(k-1)\pi(\sqrt{x})$ of them have norm $\leq x$), so infinitely many $\mathfrak{p}$ in this class have degree one: $N\mathfrak{p} = P$ is a rational prime, $P \nmid kq_1 \cdots q_t$, and $P$ splits in $K$, i.e. $P \equiv 1 \pmod{k}$. Fix such a $\mathfrak{p}$; then $O_K/\mathfrak{p} \cong \mathbb{F}_P$.

*Power residue symbols.* Let $\mathfrak{P}$ be a prime of $M$ above $\mathfrak{p}$ and $\beta = q_i^{1/k}$. By definition of the Frobenius, $\sigma_c(\beta) \equiv \beta^P \pmod{\mathfrak{P}}$, so $\zeta^{c_i}\beta \equiv \beta^P$, i.e. $q_i^{(P-1)/k} = \beta^{P-1} \equiv \zeta^{c_i} \pmod{\mathfrak{P}}$, since $\beta$ is a unit at $\mathfrak{P}$. Both sides lie in $O_K$, so the congruence holds modulo $\mathfrak{P} \cap O_K = \mathfrak{p}$.

*Definition of $f_P$.* Since $P \nmid k$, the polynomial $X^k - 1$ is separable modulo $P$, so the $k$ elements of $\mu_k$ remain distinct modulo $\mathfrak{p}$ and reduce bijectively onto the $k$-th roots of unity of $\mathbb{F}_P$ (there are exactly $k$ of them, as $k \mid P-1$). For $P \nmid n$ the element $n^{(P-1)/k} \bmod \mathfrak{p}$ satisfies $(n^{(P-1)/k})^k = n^{P-1} \equiv 1$, so there is a unique $f_P(n) \in \mathbb{Z}/k\mathbb{Z}$ with $n^{(P-1)/k} \equiv \zeta^{f_P(n)} \pmod{\mathfrak{p}}$. Property (1) follows from multiplicativity of $n \mapsto n^{(P-1)/k}$ and injectivity of reduction on $\mu_k$; property (2) is the previous paragraph. For (3): $\mathbb{F}_P^\times$ is cyclic of order $P-1$ and $k \mid P-1$, so $n$ is a $k$-th power modulo $P$ iff $n^{(P-1)/k} \equiv 1 \pmod{P}$ (Euler's criterion), iff $f_P(n) = 0$. □

*Remark* 2.3. Lemma 2.2 is classical and not new. For odd prime $k$ and primes $q_i \neq k$ it is the Lemma on p. 104 of Lehmer and Lehmer [12], stated there as a special case of a theorem of Kummer (Hilbert's *Zahlbericht* [9]); Mills [15] proved the version for arbitrary $k$, which is the statement used by Lehmer, Lehmer and Mills: "every vector $[r_1, \ldots, r_t]$ ... that satisfies certain parity conditions is the $S$-vector of infinitely many primes" [13, p. 174]; for odd prime $k$ there are no parity conditions. Graham [8] uses it in exactly the form above ("a special case of a result of Kummer"). We include a proof only because the Lean statement requires the precise hypotheses, and because we need $k$ itself among the $q_i$ ($g_5(5) = 0$, $g_7(7) = 1$), a case excluded in [12]. All known values of $\Lambda$ were obtained through this reduction.

**Corollary 2.4.** *Let $k$ be an odd prime, $g$ an assignment and $L \geq 3$. Suppose that every $f : \mathbb{N} \to \mathbb{N}$ that is completely additive mod $k$ on $[1, L]$ and realises $g$ on $[1, L]$ has no $r$ with $1 \leq r$, $r+2 \leq L$ and $f(r) \equiv f(r+1) \equiv f(r+2) \equiv 0 \pmod{k}$. Then $r(k, 3, P) \geq L - 1$ for infinitely many primes $P$; in particular $\Lambda(k, 3) \geq L - 1$.*

*Proof.* Apply Lemma 2.2 to the primes $q \leq L$ and $c_q = g(q)$, and take any of the infinitely many primes $P > L$ it provides. Define $f(n) \in \{0, \ldots, k-1\}$ as the representative of $f_P(n)$ for $P \nmid n$ and $f(n) = 0$ otherwise. Every $n \leq L$ is prime to $P$, so $f$ is completely additive mod $k$ on $[1, L]$ and realises $g$ there. By hypothesis no $r \leq L - 2$ has $r, r+1, r+2$ all $k$-th power residues modulo $P$, so $r(k, 3, P) \geq L - 1$. □

## 3 The assignments

**Definition 3.1.** The assignment $g_5$ (values in $\mathbb{Z}/5\mathbb{Z}$) is $g_5(q) = 1$ for every prime $q$ except the 25 primes in Table 1. The assignment $g_7$ (values in $\mathbb{Z}/7\mathbb{Z}$) is $g_7(q) = 1$ for every prime $q$ except $g_7(2) = g_7(47) = g_7(163) = 0$.

For an assignment $g$ write $\chi_g(n) = \sum_q v_q(n)\, g(q) \in \mathbb{Z}/k\mathbb{Z}$ for its completely additive extension. With $g \equiv 1$ (the "$\Omega$-assignment") $\chi_g(n) = \Omega(n) \bmod k$; its first triple of consecutive zeros is at 15470 for $k = 5$ and at 3145310 for $k = 7$. The assignments $g_5, g_7$ were found by a local search (Section 5) that starts from the $\Omega$-assignment and changes the values of a few primes until no zero triple remains below a target bound.

| $q$ | 2 | 5 | 7 | 11 | 13 | 17 | 23 | 31 | 73 | 347 | 409 | 617 | 1033 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $g_5(q)$ | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 2 | 2 | 2 | 2 |
| $q$ | 1373 | 2053 | 2089 | 2099 | 2437 | 2621 | 3907 | 4523 | 7057 | 12917 | 21407 | 49613 | |
| $g_5(q)$ | 0 | 0 | 2 | 2 | 2 | 0 | 2 | 0 | 2 | 2 | 3 | 3 | |

Table 1: The exceptional primes of $g_5$; all other primes have $g_5(q) = 1$.

# 4 The Lean-certified core

**Theorem 4.1** (certified in Lean)**.**

1. *Let $L = 10^8$. Every $f : \mathbb{N} \to \mathbb{N}$ that is completely additive mod 5 on $[1, L]$ and realises $g_5$ on $[1, L]$ satisfies: there is no $r$ with $1 \le r$, $r + 2 \le L$ and $f(r) \equiv f(r + 1) \equiv f(r + 2) \equiv 0 \pmod 5$.*
2. *The same holds for $L = 10^7$, modulus 7 and $g_7$.*

Theorem 1.1 follows from Theorem 4.1 and Corollary 2.4.

## 4.1 The Lean statement

The file `lean/Main.lean` (Lean 4.34.1, core library only, no imports) contains, verbatim (docstrings omitted):

```
def L : Nat := 100000000
def IsPrime (p : Nat) : Prop := 2 ≤ p ∧ ∀ d, d ∣ p → d = 1 ∨ d = p
def g (p : Nat) : Nat :=
  if 49613 < p then 1 else
  if p = 2 then 0 else if p = 5 then 0 else if p = 7 then 0 else if p = 11 then 0 else
  if p = 13 then 0 else if p = 17 then 0 else if p = 23 then 0 else if p = 31 then 0 else
  if p = 73 then 0 else if p = 347 then 2 else if p = 409 then 2 else if p = 617 then 2 else
  if p = 1033 then 2 else if p = 1373 then 0 else if p = 2053 then 0 else if p = 2089 then 2
    else
  if p = 2099 then 2 else if p = 2437 then 2 else if p = 2621 then 0 else if p = 3907 then 2
    else
  if p = 4523 then 0 else if p = 7057 then 2 else if p = 12917 then 2 else if p = 21407 then 3
    else
  if p = 49613 then 3 else 1

theorem no_three_consecutive_quintic_residues (f : Nat → Nat)
    (hmul : ∀ a b, 1 ≤ a → 1 ≤ b → a * b ≤ L → f (a * b) % 5 = (f a + f b) % 5)
    (hprime : ∀ p, p ≤ L → IsPrime p → f p % 5 = g p % 5) :
    ∀ r, 1 ≤ r → r + 2 ≤ L → ¬ (f r % 5 = 0 ∧ f (r + 1) % 5 = 0 ∧ f (r + 2) % 5 = 0)
```

`lean/Main7.lean` is the same file with modulus 7, `L := 10000000`, `B := 3163` and

```
def g (p : Nat) : Nat :=
  if 163 < p then 1 else
  if p = 2 then 0 else if p = 47 then 0 else if p = 163 then 0 else 1
```

and main theorem `no_three_consecutive_septic_residues`.

## 4.2 Proof architecture

The proof introduces a table and a bound that do *not* occur in the statement:

- `T : ByteArray := buildT L`, an array of $L+1$ bytes in which a sieve stores $T[n] = \sum_{p^j \mid n} g(p) \bmod 5$, i.e. $\chi_g(n)$. The sieve is *not trusted*: only the properties below, checked on the computed array, are used.
- `B := 10000`, with $L < (B+1)^2$.

A single Boolean `bigCheck` verifies, for the computed table:

(i) $T[1] \equiv 0$ and $L < (B+1)^2$;

(ii) for every $p \le B$ that has no divisor $e$ with $2 \le e$, $e^2 \le p$ (a trial-division primality test) and every $2 \le m \le L/p$: $T[pm] \equiv T[p] + T[m]$;

(iii) for every $n \in [2, L]$: $T[n] \equiv g(n)$, or $n$ has a divisor $e$ with $2 \le e$, $e^2 \le n$;

(iv) there is no $r \in [1, L-2]$ with $T[r] \equiv T[r+1] \equiv T[r+2] \equiv 0$.

The theorem `bigCheck_true : bigCheck = true` is proved by `native_decide`. All remaining steps are kernel-checked:

- soundness lemmas turn `bigCheck = true` into the Propositions (i)–(iv) (`loopAll_sound`, `pairLoop_sound`, `hsd_sound`, `quickComp_sound`);
- `agree`: by strong induction, $f(n) \equiv T[n]$ for all $n \in [1, L]$. For $n = 1$, $f(1) \equiv f(1) + f(1)$ gives $f(1) \equiv 0 \equiv T[1]$. If $n$ is prime, (iii) applies with its second alternative impossible (a divisor $e$ with $2 \le e$, $e^2 \le n$ would be a proper divisor), so $T[n] \equiv g(n) \equiv f(n)$. If $n \ge 2$ is not prime, write $n = ab$ with $2 \le a \le b$ (`split_composite`), take a prime $p \mid a$ (`exists_prime_dvd`); then $p^2 \le a^2 \le n \le L$, so $p \le B$ by (i), $p$ passes the primality test in (ii), and with $n = pm$ we have $m \ge p \ge 2$, whence $f(n) \equiv f(p) + f(m) \equiv T[p] + T[m] \equiv T[n]$ by `hmul`, the induction hypothesis and (ii);
- the main theorem then follows from (iv).

### 4.3 Axioms, the use of `native_decide`, and runs

`#print axioms` reports, for both main theorems,

```
[propext, Classical.choice, Quot.sound, bigCheck_true._native.native_decide.ax_1_1]
```

and for `bigCheck_true` itself

```
[propext, Quot.sound, bigCheck_true._native.native_decide.ax_1_1].
```

There is no `sorry`, `admit` or user-declared axiom. The last axiom expresses trust in Lean's evaluator for the single closed Boolean term `bigCheck`: the trusted base of Theorem 4.1 is the Lean kernel *plus* the compiler/IR interpreter invoked by `native_decide` (`Lean.ofReduceBool`). A reader who does not extend trust to the latter should regard Theorem 4.1 as computer-checked rather than kernel-checked; the two independent C programs of Section 5 confirm the same table-level facts.

**Why not the kernel.** `bigCheck` builds an array of $10^8 + 1$ bytes by a sieve and then performs of the order of $10^9$ table look-ups (checks (ii)–(iv)). The Lean kernel evaluates terms by reduction, without a constant-time array representation; even the interpreter needed about 82 minutes for $L = 10^8$. A kernel reduction of this size is far out of reach, so the check is delegated to `native_decide`, and only to this one closed term.

**Runs.** Lean 4.34.1 on a shared 14-core machine: `Main.lean` ($k = 5$, $L = 10^8$): wall time 4940 s, 725 MB; `Main7.lean` ($k = 7$, $L = 10^7$): 165 s, 540 MB. Both were run with plain `lean <file>` (no project file, no Mathlib) under a timing/memory-limit wrapper; `Main7.lean` with the extra flag `-j 2`. An independent re-run by the reviewer with the same toolchain reproduced both outputs (see `README.md`). Two control files (in `lean/`) show that the check is not vacuous: with $g \equiv 1$ the same `bigCheck` evaluates to `false` for $L = 15500$ (zero triple 15470–15472) and to `true` for $L = 15469$. An earlier version of the checker, organised differently but with the same proof plan, certified the $k = 5$ statement for $L = 10^7$.

**Statement audit.** The hypotheses quantify over *all* $f : \mathbb{N} \to \mathbb{N}$; `hmul` is complete additivity mod 5 on $[1, L]$ (all $a, b \geq 1$ with $ab \leq L$), and `hprime` prescribes $f$ on every prime $\leq L$ via `IsPrime`, the textbook definition. These are exactly the properties of the index functions $f_P$ of Lemma 2.2 for $P > L$. The function `g` is a plain if-chain whose values agree with Definition 3.1; the leading test `49613 < p` only short-cuts the chain (no exceptional prime exceeds 49613). The conclusion covers every window $r, r+1, r+2$ contained in $[1, L]$, so a first zero triple must start at $r \geq L - 1$, as used in Corollary 2.4. The table `T`, the bound `B` and the sieve do not appear in the statement.

## 5 Computational results (not formally certified)

**Proposition 5.1** (computation)**.** *The completely additive extension $\chi_{g_5}$ has no triple of consecutive zeros $r, r+1, r+2$ with $r + 2 \leq 313\,714\,025$, and*

$$\begin{aligned} 313714024 &= 2^3 \cdot 13^4 \cdot 1373, & \chi_{g_5} &= 3 \cdot 0 + 4 \cdot 0 + 0 = 0, \\ 313714025 &= 5^2 \cdot 29^2 \cdot 43 \cdot 347, & \chi_{g_5} &= 0 + 2 + 1 + 2 \equiv 0, \\ 313714026 &= 2 \cdot 3^3 \cdot 997 \cdot 5827, & \chi_{g_5} &= 0 + 3 + 1 + 1 \equiv 0 \pmod 5. \end{aligned}$$

*Likewise $\chi_{g_7}$ has no zero triple with $r + 2 \leq 180\,317\,501$, and its first zero triple is* $180317500 = 2^2 \cdot 5^4 \cdot 11 \cdot 79 \cdot 83$, $180317501 = 7^4 \cdot 13 \cdot 53 \cdot 109$, $180317502 = 2 \cdot 3^4 \cdot 37 \cdot 67 \cdot 449$.

By Corollary 2.4 (with $L = 313\,714\,025$, resp. $L = 180\,317\,501$), Proposition 5.1 gives $\Lambda(5, 3) \geq 313\,714\,024$ and $\Lambda(7, 3) \geq 180\,317\,500$; moreover $r(5, 3, P) = 313\,714\,024$ for infinitely many $P$, namely the primes of Lemma 2.2 for the primes $q \leq 313\,714\,026$.

**How this was checked.** Two independent C programs compute $\chi_g$ and scan for zero triples: one uses a smallest-prime-factor table and the recursion $\chi(n) = g(p) + \chi(n/p)$; the other factors each segment of $2^{24}$ integers by trial division by all primes up to $\sqrt{L}$. Both find the same first triples. The three factorisations above were also checked with SymPy and can be verified by hand. These computations are not formally certified. `Main.lean` with `L := 313714025` and `B := 17713` (so that $(B+1)^2 > L$) would certify the bound $\Lambda(5, 3) \geq 313\,714\,024$, but we did not run it: at the measured interpreter speed it would need several hours.

**Search.** The assignments were found by a WalkSAT-type local search over the values of the primes, with incremental bookkeeping of zero triples: a zero window is picked at random and the value of one of the primes dividing its members is changed so as to minimise the number of zero windows. Starting from the $\Omega$-assignment, 4 and 36 moves removed all zero triples up to $10^6$ and $10^7$ ($k = 5$); 61 further moves extended a $10^7$ solution to $10^8$; for $k = 7$, three moves sufficed up to $10^8$. An exact depth-first search with conflict-directed backjumping over the same model reproduces the known values $\Lambda(2, 2) = 9$, $\Lambda(3, 2) = 77$, $\Lambda(4, 2) = 1224$ and $\Lambda(5, 2) = 7888$ as the

boundary between solvable and unsolvable bounds, which checks the model (including Lemma 2.2) against the literature for $k = 2, 3, 5$. (For $k = 4$ this is only a plausibility check: Lemma 2.2 is stated for odd prime $k$, and the symmetry reduction used by the search—scaling all values by a unit of $\mathbb{Z}/k\mathbb{Z}$, so that the first non-zero prime value may be taken to be 1—is transitive on the non-zero values only for prime $k$; for $k = 4$ the search therefore omits assignments whose first non-zero value is 2, and its "unsolvable" verdict at 1225 is not a complete search.)

**Side data.** The $\Omega$-assignment alone gives $\Lambda(11, 2) \geq 6\,082\,047$ (first zero pair $6082047 = 3^{10} \cdot 103$, $6082048 = 2^9 \cdot 7 \cdot 1697$) and $\Lambda(13, 2) \geq 10^8$ (no two consecutive integers below $10^8$ have $\Omega \equiv 0 \pmod{13}$), again by computation only. These two bounds are *weaker* than what has been known since 1964: Brillhart, Lehmer and Lehmer [4, §§1, 6] proved $\Lambda(k, 2) > 5^k$ for every odd $k \geq 5$ by an elementary argument (the assignment $R(3) = 0$, $R(q) = 1$ for $q \neq 3$, and factorisations of $3^n \pm 1$), so $\Lambda(11, 2) > 48\,828\,125$ and $\Lambda(13, 2) > 1.2 \cdot 10^9$; they remark that $7 \cdot 5^{k-1}$ can be obtained similarly, and they give $\Lambda(8, 2) \geq 1\,200\,744$, $\Lambda(9, 2) > 10^7$, $\Lambda(10, 2) \geq 22\,458\,303$. Elementarily, for every odd prime $k$ the $\Omega$-assignment shows $\Lambda(k, 2) \geq 3^k - 1$ and $\Lambda(k, 3) \geq 3^k - 2$: every member $n \geq 2$ of a zero run has $\Omega(n) \geq k$, and one member is odd, hence at least $3^k$. We claim no novelty for these remarks.

## 6 Finiteness remains open

We stress that Theorem 1.1 and Proposition 5.1 are lower bounds; they do not decide whether $\Lambda(5, 3)$ or $\Lambda(7, 3)$ is finite.

By Lemma 2.2, $\Lambda(k, 3)$ is finite if and only if for some $L$ every assignment $g$ has a zero triple of $\chi_g$ in $[1, L]$ (if not, Corollary 2.4 gives $\Lambda(k, 3) \geq L - 1$ for every $L$; conversely every prime $P \equiv 1 \pmod{k}$ induces an assignment). We found no sign of such an $L$: the exact search slows down near $1.4 \cdot 10^5$ without proving unsolvability, while the local search finds solutions far beyond with very few changes.

The method behind $\Lambda(3, 3) = 23532$ [14] and the values of $\Lambda(k, 2)$ restricts to the first $t$ primes and uses runs of consecutive $p_t$-smooth integers, whose zero conditions are linear forms on $(\mathbb{Z}/k\mathbb{Z})^t$. By the analogue for triples of the heuristic count in [13, p. 174], covering $(\mathbb{Z}/5\mathbb{Z})^{25}$ by the codimension-three subspaces coming from triples would need about $-t \log 5 / \log(1 - 5^{-3}) \approx 5000$ triples of consecutive 97-smooth integers, while there are only 787 such triples below $10^{12}$ (our count). So the approach of 1962 does not extend in an obvious way. Nor can periodic constructions prove $\Lambda(k, 3) = \infty$ for odd $k$: if $\chi$ agrees on integers prime to $M$ with a character $\psi$ of $(\mathbb{Z}/M\mathbb{Z})^\times$ of order dividing $k$ (written additively, with values in $\mathbb{Z}/k\mathbb{Z}$), then $\psi(-1) = 0$ as $k$ is odd, so $M^k - 1$ and $M^k + 1$ (both $\equiv \pm 1 \pmod{M}$) are zeros of $\chi$, and $\chi(M^k) = k\,\chi(M) = 0$; thus $(M^k - 1, M^k, M^k + 1)$ is a zero triple.

## 7 Data and reproduction

All files are in the directory of this preprint.

- Lean files `lean/Main.lean` and `lean/Main7.lean`, with outputs `lean/lean_output.txt` and `lean/lean_output_k7.txt`; control files `lean/NegOpt.lean` and `lean/PosOpt.lean`. Check with `lean Main.lean` run inside `lean/` (Lean 4.34.1, core library only; no `lakefile` is needed).

- `data/assignment_k5.txt`, `data/assignment_k7.txt`: the exceptional primes and their values (all other primes have value 1).
- `scripts/`: C sources of the local search (`ws2.c`), the exact search (`lam.c`), the two independent checkers (`vchk.c`, `segchk.c`), the $\Omega$-baseline (`om.c`), and `reproduce.sh`.

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